Anhad Mehrotra

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EDUCATION

Cluster Innovation Centre, University of Delhi Nov. 2022 – May 2026
B.Tech in Information Technology and Mathematical Innovations New Delhi, India

- **CGPA:** 9.16 / 10

EXPERIENCE

Edxso Feb. 2026 – Apr. 2026
AI Engineer Intern New Delhi, India

- Built an AI assessment platform (Gemini API, Docker) that evaluates question papers against NCF competency frameworks and generates alignment reports, with role-based access control.
- Developed and deployed a GenAI pipeline on DigitalOcean for automated product visual creation supporting single and batch workflows.
- Contributed to a scalable question bank system scoring submissions against defined competencies with a reward mechanism to incentivize quality contributions.

National e-Governance Division, Ministry of Electronics & IT Jun. 2025 – Aug. 2025
AI/ML Intern New Delhi, India

- Developed multilingual semantic search and feature enhancements for 3600+ schemes on the myscheme portal and mobile app using FAISS and SentenceTransformers.
- Built an agentic AI-powered course recommender with FastAPI, integrating hybrid filtering with semantic search and multi-stage re-ranking. 7 course provider platforms covered.
- Researched AI security threats, governance frameworks, and operational measures including API security and red teaming.

Craving For Gaming Sep. 2024 – Feb. 2025
Web Developer Intern New Delhi, India

- Optimized website performance and improved UX using HTML/CSS and on-page SEO techniques.
- Supported a product listing revamp and improved user engagement by 18%.

PROJECTS

Multi-Agent Research & News Automation Bot Oct. 2026

- Built a **multi-agent pipeline** to autonomously discover, filter, and summarize AI research papers and domain-specific news, delivering daily digests via Telegram.
- Designed a **reusable BaseAgent abstraction** and topic-specific agents, enabling extensibility through **inheritance and separation of concerns**.
- Implemented a **global semaphore-based rate limiter** to serialize concurrent LLM requests; added **cross-agent URL deduplication**, retry logic with exponential backoff, and automatic Telegram message chunking for reliable delivery. Github

LLM-based Opinion Dynamics Simulation Apr. 2026

- Built a **multi-agent simulation framework** using LLMs to model opinion formation and evolution in social systems under varying influence, bias, and network structures.
- Analyzed emergent behaviors including **polarization, consensus formation, and opinion clustering** across different simulation settings. Github

Nanoparticle Toxicity Prediction Using Deep Learning Nov. 2025

- Designed a custom deep neural network for cross-dataset nanotoxicity prediction; benchmarked against a Random Forest baseline on a general nanotoxicity dataset.
- **Improved cross-dataset accuracy from 75% (RF) to 86.75%** on a lung macrophage dataset.
- Applied PDP, 2D-PDP, and SHAP analyses to identify dominant toxicity factors. Github

ACHIEVEMENTS

BioMed Bharat Hackathon 2026 | *Winner / 200+ teams [Certificate]* Sep. 2026

- Built an **intelligent hospital laboratory monitoring platform** integrating EMR, LIS, and HIS via **HL7 FHIR**, training **XGBoost on MIMIC-III** for temporal and cross-parameter trend analysis with **79% test accuracy** and **81.5% sensitivity**, alongside explainable clinical alerts delivered via mobile app and a custom pocket-sized hardware. Overview

TATA Technologies InnoVent Hackathon | *Top 31 / 2,800+ teams [Certificate]* Aug. 2025

- Built an AI-driven EV battery analytics platform with transformer- and RNN-based models for RUL estimation, achieving **85%+ accuracy**; dual-layer architecture for data processing and API delivery. GitHub

Hack The Future Hackathon, MoSPI | *Top 20 / 1,000+ teams [Certificate]* Mar. 2025

- Built a multilingual semantic search app for NIC code classification with voice-to-text and synonym mapping; **improved relevance by 21%** over baseline. GitHub

PUBLICATIONS

Design of an Optimal Planning Framework for Cryosurgical Treatment of Brain Tumor using CNN Segmentation of MRI Images | *Cryobiology, Elsevier* 2026

- Co-authored research integrating U-Net based MRI segmentation pipeline (Dice 0.872) with optimization algorithms (K-Medoids, GMM, BPA) for cryosurgical planning.
- Built a simulation framework in COMSOL Multiphysics using the Pennes Bioheat Transfer Equation for thermal modeling of cryoablation.
- **DOI:** https://doi.org/10.1016/j.cryobiol.2026.105619

SKILLS

Languages : Python, C, Java, SQL
ML / DL : TensorFlow, Keras, PyTorch, scikit-learn, OpenCV, NLTK, FAISS, SentenceTransformers
GenAI : LangChain, LangGraph, CrewAI, RAG, MCP, agentic workflows, prompt engineering, embeddings
Data : NumPy, Pandas, SciPy, Power BI
Tools : Git, Docker, FastAPI, Linux, DigitalOcean, MATLAB, Arduino, n8n, COMSOL Multiphysics, Jupyter, Colab
Databases : PostgreSQL, MongoDB, Redis
Web : HTML, CSS, React.js, Streamlit